

Power Plant Engineering

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Depreciation-Straight line method

Annual depreciation reserve= $(P-S)/n$

where

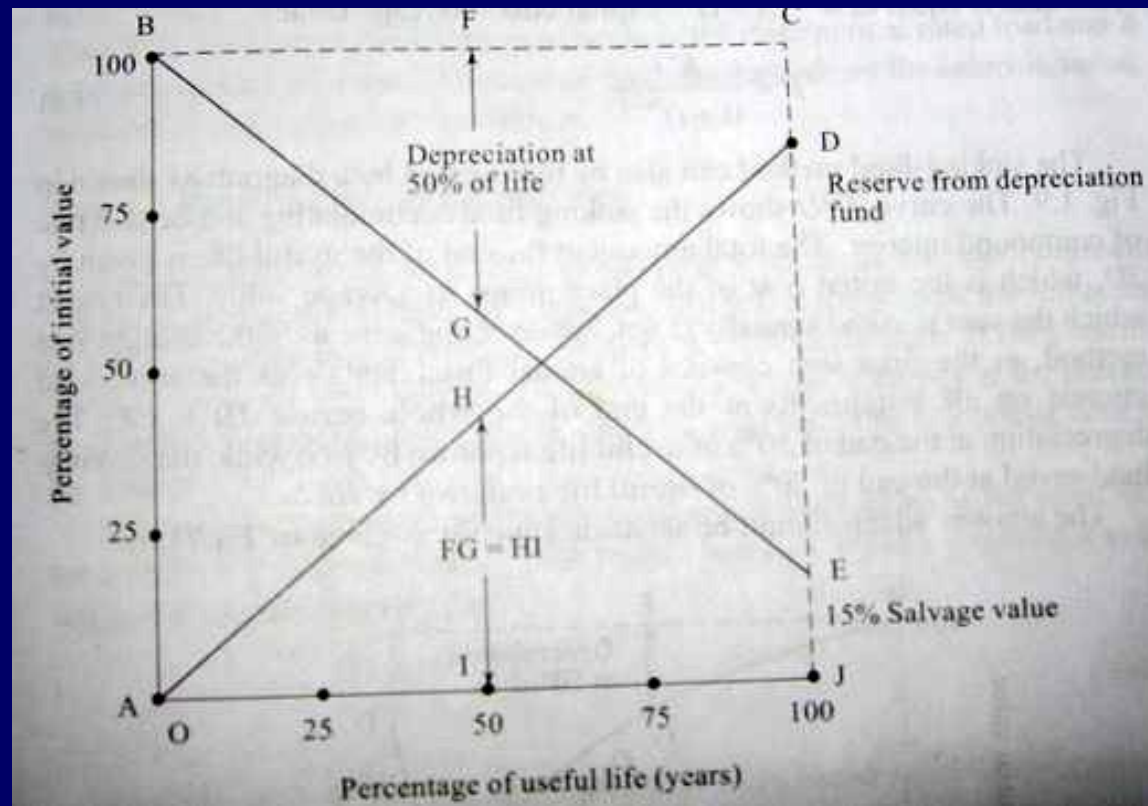
P = Principal ,ie capital cost of the plant

S = Salvage value or Residual value of power plant at the end of years.

i = Probable duration of useful life of power plant in years.

The amount collected every year as depreciation fund does not depend on the interest it may draw. Any interest if earned by the depreciation amount is considered as income.

Depreciation – Straight line method



Sinking Fund Method

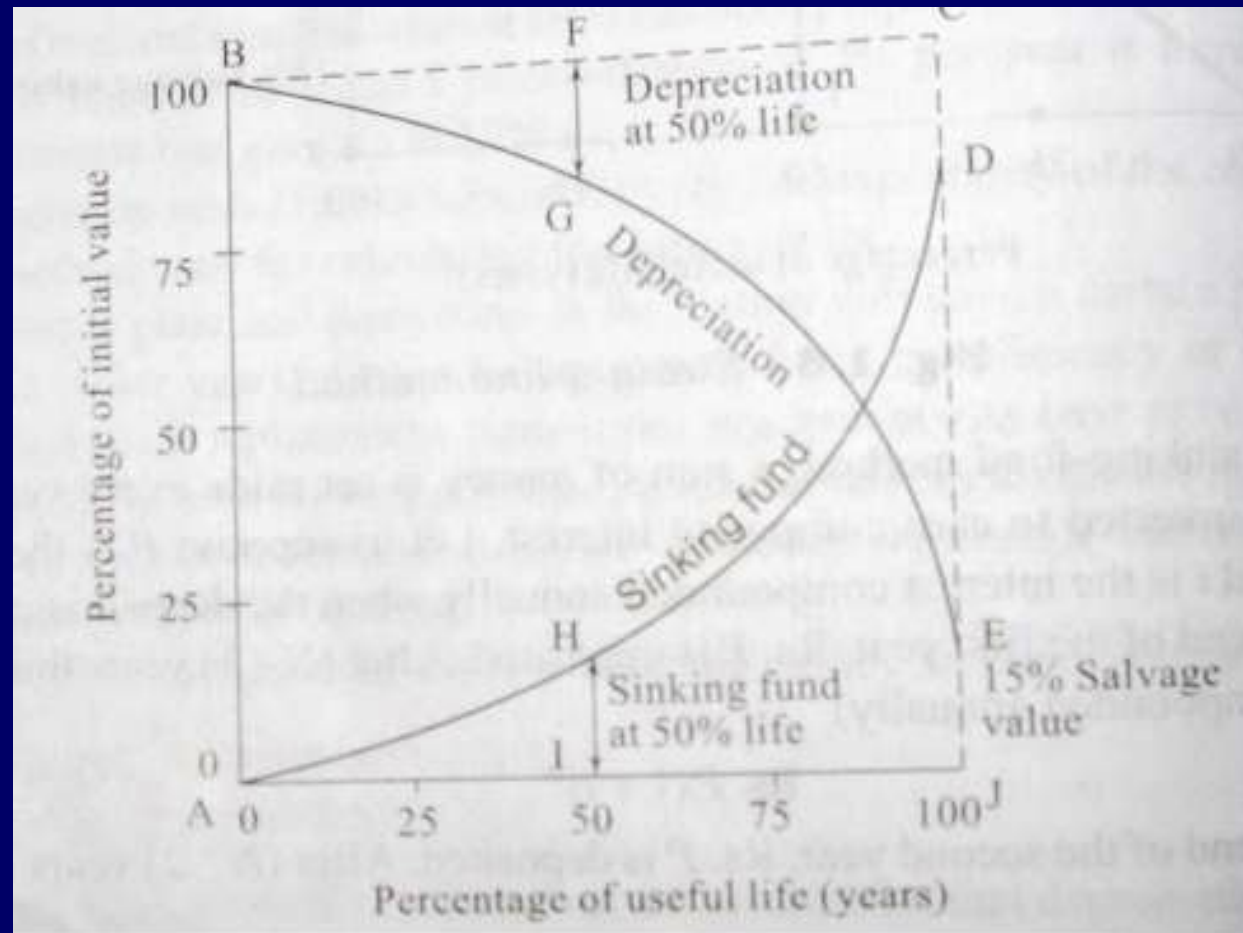
In this method the annual amount to be set aside per year consist of annual instalments plus the interest on all the instalments.

Annual depreciation reserve

$$= (P - S) [r/(1 + r)^n - 1]$$

where r = rate of compound interest.

Depreciation – Sinking fund method



In a steam power plant the capital cost of power generation equipment is Rs. 2000000. The useful life of the plant is 30 years and salvage value of the plant to Rs. 100000. Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 6%.

$P = \text{Capital cost} = \text{Rs. } 2000000$

$S = \text{Salvage value} = \text{Rs. } 100000$

$n = \text{Useful life} = 30 \text{ years}$

$r = \text{Compound interest} = 6\%$

$A = \text{Amount to be saved per year for replacement}$

$$= (2000000 - 100000) * .06 / ((1 + .06)^{30} - 1)$$

$= \text{Rs. } 24,000. \text{ Ans.}$

Straightline method

$$\begin{aligned}\text{Yearly depreciation} &= (2000000 - 100000) / 30 \\ &= 63333\end{aligned}$$